Attributing annotator polarization to sociodemographic groups

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1 Introduction

Annotations are essential for a wide range of tasks, especially in fields such as content moderation, toxicity detection and hate speech detection. These tasks are especially essential for training systems that protect vulnerable and minority groups in online spaces, where they are often targeted [20, 5]. However, annotations for such tasks are subjective and usually based on majority-group annotators. This may at best limit the usefulness of the data, or at worst lead to biased and misconfigured systems.

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree in vain. An example of such cases would be racist comments that are presented with coded language (usually referred to as "dog-whistles" [14]), which are not picked up by most annotators, but only by ones belonging in the targeted minority. Inversely, majority-group annotators may bias their own annotations against the minority-group annotators. Sap et al. [17] for instance show that toxicity models disproportionally marked African American speech as "toxic".

Inter-annotator agreement is often used to detect such cases [9]. Metrics based on agreement mostly measure annotation variance, which may not be an optimal problem formulation for detecting such cases. *Polarization* is a better instrument in this regard, since it formulates annotation analysis as a cluster identification problem [12, 13]—specifically whether two or more annotation clusters are present (Figure 1).

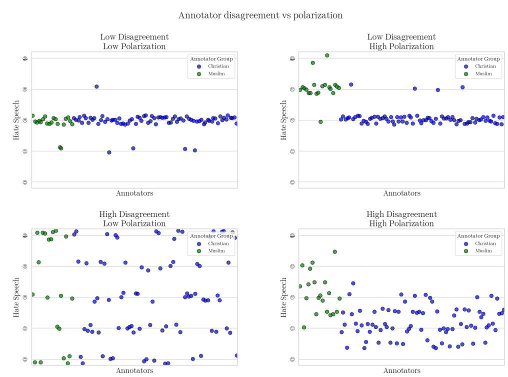

Figure 1: Disagreement is similar to variance; thus it overlooks cases where annotators may be "split down the middle". Polarization on the other hand, is able to identify multiple clusters even if they are close together.

While we can detect polarization [13, 12], there is little work on how to detect whether it is actually caused by marginalized groups. Pavlopoulos and Likas [13] propose a mechanism which only works in the extreme case where overall polarization is zero; in this work we demonstrate that this case is not frequent in real-world data. We instead propose the *"Aposteriori Unimodality (Apunim) Metric"*, which attributes polarization to individual annotator sociodemographic groups. Our contributions are:

1. We introduce a new metric which measures how much polarization can be attributed specific sociodemographic groups.

2. We discover issues not yet acknowledged in literature, such as the presence of inherent polarization and conflicting polarization directions. Our

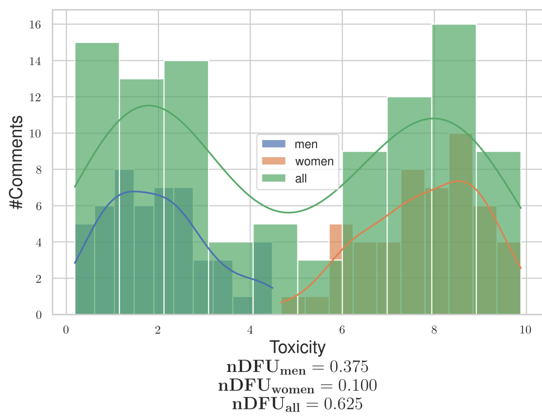

Figure 2: Example of a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

proposed metric is designed to avoid these issues.

3. We apply our metric to four datasets; two with human comments and annotations [16, 10] as well as two generated and annotated by Large Language Model (LLM) agents [19].

2 Methodology

2.1 Problem Formulation

SocioDemographic Backgrounds Since our goal is to pinpoint which specific characteristics contribute to polarization, we need to isolate individual groups within a SocioDemographic Background (SDB). Let Θ be one of multiple "dimensions" the set of all annotator SDB groups for a single dimension (e.g. "male", "female" and "non-binary" if we are investigating annotator gender).

Annotations Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated data points. We assume that annotating a data point depends on two variables: inherent ("apriori") controversy, and the ("aposteriori") annotator's SDB, as well as uncontrolled factors such as mood and personal experiences (noise). Assuming that each data-point c is assigned multiple annotators, we can define the annotation multi-set $A(c) = \{(a, \theta)\}$, where a is a single annotation. As an example, the annotations for a comment in a sentiment analysis task would be formulated as:

$$
\begin{aligned}
&A(\textit{"Could be better, could be worse"}) = \\
&\{(positive,\, female),\, (negative,\, male), \\
&(positive,\, female),\, \ldots\}
\end{aligned}
\tag{1}
$$

Polarization Each set of annotations features a certain degree of *polarization*. Unpolarized annotations are usually unimodal; the annotators disagree on the details (which would be shown roughly as a bell curve around the median annotation), not fundamentally (which would likely be shown as a multimodal distribution). Pavlopoulos and Likas [13] create a polarization metric, the "normalized Distance From Unimodality (nDFU)", which measures whether an annotation set shows no polarization (unimodal distribution—$nDFU \approx 0$), up to complete polarization (multimodal distribution—$nDFU \approx 1$). Thus, we need a metric that measures how much the nDFU of a polarizing data point can be partially explained by θ.

2.2 Intuition

Comment polarization Intuitively, θ partially explains the polarization of a data point c when the annotations grouped by θ are more polarized compared to the full set of annotations. Figure 2 exhibits a hypothetical example where a misogynistic comment is annotated for toxicity by male and female annotators.[1] The annotations are generally polarized ($nDFU_{all} = 0.625$). The annotations by female annotators exhibit low polarization between themselves ($nDFU_{women} = 0.1$), since most agree the data-point is toxic. The set of male annotations, also shows low polarization ($nDFU_{men} = 0.3725$), but for the opposite reason—most men agree that it is *not* toxic. This suggests that the overall polarization is driven

[1] The use of only two predominant genders is made only for the purposes of demonstration.

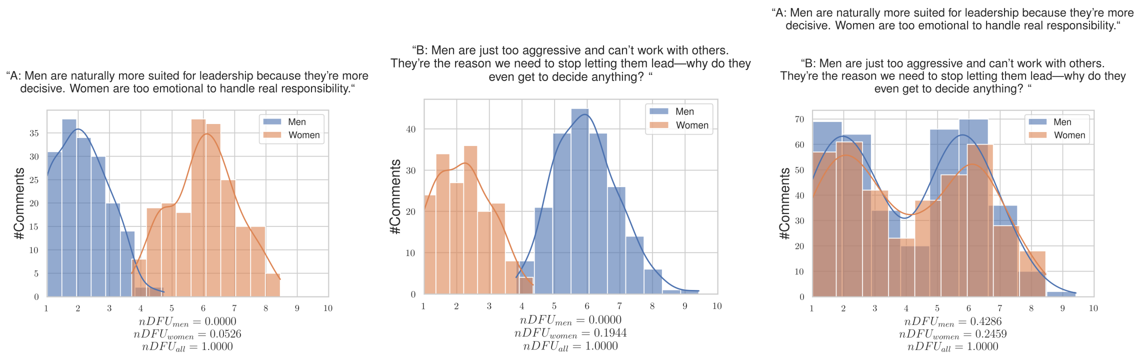

Figure 3: Hypothetical example of a polarizing discussion with two comments, for which the annotators disagree on which is the toxic one. If we aggregate the two comments, the polarization scores for both men and women significantly rise, obscuring whether the exhibited polarization can be partially attributed to gender.

by disagreements between male and female annotators.

Dataset polarization Given this observation, we would be tempted to aggregate all annotations in the dataset and compare the polarization of each θ with the full set of annotations $A(c)$. In fact this is very tempting since aggregating annotations solves the very real and important problem of annotation scarcity; each data point may be annotated by only 5 annotators, leading to a best-case 3-2 split between groups. Since nDFU is at its core a histogram estimation, 3 annotations may not be enough to acquire a reliable histogram (this is discussed in depth in §**??**).

However, this formulation would not work well. Figure 3 illustrates a hypothetical discussion with two comments, both of which are toxic, but where male and female annotators disagree on *which* comment is the toxic one. If we aggregate the annotations to a single discussion, the opposing polarization effects might balance each other out, leading to a false negative. This example demonstrates that polarization has a *direction*, which is not captured by the (symmetric) $nDFU$ metric.

2.3 Aposteriori polarization

The observed polarization of a data-point on the group of annotations θ can be directly computed as:

$$pol(c, \theta) = nDFU(P(A(c), \theta) \tag{2}$$

where $P(A(c), \theta_i) = \{a(c; \theta) \in A(c) | \theta = \theta_i\}$ is the partition of A for a data-point c, for which its annotators belong to the SDB group θ_i.

In §1 we mentioned that annotator polarization can be caused either by the annotators SDB, or other factors. The latter can be considered "apriori" polarization, which is driven by the inherent "controversy" of the data point. Previous work [13] considered the case where this apriori polarization was zero; this is not the case in many datasets, as we demonstrate in §3.

Given infinite annotations, the assumption of independence would hold. Likewise, if we compared the subset with infinite random subsets of the superset. We can thus estimate the apriori polarization in data point c by bootstrapping (Figure 8). We randomly partition the annotations in $|\Theta|$ groups, with matching group sizes (e.g., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we will create random partitions of sizes 80 and 20). The randomly partitioned annotations

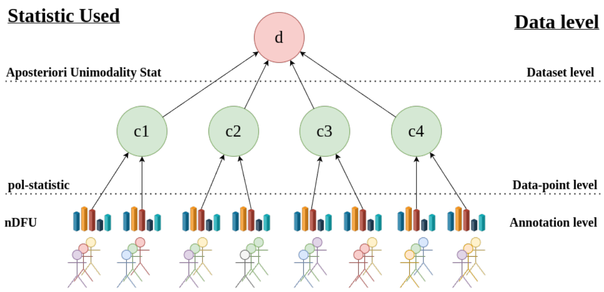

Figure 4: An overview of the Aposteriori Unimodality Test. We gather statistical information from the annotations (different SDBs are denoted by different colors) via the nDFU measure, which is aggregated on the data-point-level by the pol-statistic. The Aposteriori Unimodality Test is applied on the discussion level, operating on the pol-statistics of the individual data-points.

are then sampled t times. Formally, the estimated apriori polarization will be given by:

$$\frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{3}$$

where $\tilde{P}_i$ is the random partition operator with matching group sizes.

While data scarcity is an important limitation when acquiring the observed polarization, this is decisively not the case with the number of data points themselves, which can range up to hundreds of thousands. Thus, we have the luxury of filtering data points before computing a dataset-wide metric. Since our metric attributes polarization, unpolarized data points are useless. The same applies for data points which are composed of only a single annotator group. Thus, we define the subset S_d of dataset d such that:

$$\begin{aligned} S_d = \{c \in d \mid nDFU(c) > \alpha \\ \textit{and } c \textit{ annotated by at least two groups}\} \end{aligned} \tag{4}$$

where α is a small positive number (in our experiments $\alpha = 0.2$).

By obtaining the pol-statistics for all data-points in S_d for SDB θ, we can get the mean observed polarization for the dataset:

$$pol_{obs.}(d,\, \theta) = \frac{1}{|d|}\sum_{c \in S_d} pol(c,\, \theta) \tag{5}$$

and similarly the mean apriori dataset polarization:

$$pol_{apriori}(d) = \frac{1}{|d|}\sum_{c \in S_d}\frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{6}$$

We can then define the *"Aposteriori Unimodality" (Apunim)* Statistic as:

$$apunim(d,\, \theta) = \frac{pol_{obs.}(d,\, \theta) - pol_{apriori}(d)}{1 - pol_{apriori}(d)} \tag{7}$$

Since $nDFU \to [0,\, 1] \Rightarrow apunim \to [-1,\, 1]$. If $apunim(d,\, \theta) \approx 0$, the exhibited polarization among annotators of group θ does not surpass what would be expected by chance. If $apunim(d,\, \theta) > 0$, then the group partially explains a rise in polarization, and inversely for the case $apunim(d,\, \theta) < 0$. The scope of the statistic and its relationship with the previously presented statistics are demonstrated in Figure 4.

2.4 p-value estimation

By obtaining the pol-statistics for all data-points in a dataset d we can apply any statistical means test with the null hypothesis:

$$H_0 : \forall \theta \in \Theta,\ apunim(d,\, \theta) \leq 0 \tag{8}$$

versus the (multiple) alternative hypotheses:

$$H_{\theta_i} : apunim(d,\, \theta_i) > 0 \tag{9}$$

Our test assumes that (1) there exist enough data-points to reliably run the means-test, and (2) enough annotations in each data-point and SDB group to estimate the *pol-statistics*. We discuss the former assumption in Appendix C. The latter can be safely assumed for most datasets, because they feature enough data-points to reliably assume normal residuals ($n \gg 30$—see §3), which is why we use the Student-T test [18] .

Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values. We choose the Holms

Table 1: Brief overview of the datasets used in this study. We are interested in the number of comments (#Comm.), annotators per comment (#Ann.) and number of SDB dimensions (#Dims.).

Name	# Comm.	# Ann.	# Dims
Kumar et al. [10]	106,035	5-6	10
Sap et al. [16]	626	5-6	4
DICES-350 [2]	72,103	60-70	2
DICES-990 [2]	43,050	123	2

method [7]. The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [4]). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

3 Datasets

We use the datasets provided by Kumar et al. [10] and Sap et al. [16]. They both feature various online comments, each annotated by a number of human annotators (5 and 4-6 annotators per comment respectively). The Kumar et al. [10] dataset stands out for featuring more than 100,000 comments, while also including 10 different dimensions for SDBs and annotator experiences.

We also use the DICES-350 and DICES-990 datasets [2], which are concerned with LLM safety. They feature less annotated comments, but more than an order of magnitude more annotators per comment, making them an invaluable resource for our study.

Brief dataset characteristics are given in Table 1. There is a reasonable amount of polarization in most annotations for all datasets as demonstrated by Figure 5.

Table 2: Aposteriori unimodality results for the dices-350 dataset.

SDB Feature		apunim
Age	1) Gen. X+	0.0521***
	2) Millenial	-0.0090
	3) Gen. Z	-0.0259
Education	College degree or higher	0.0308
	High school or below	-0.0581***
	Other	-0.0006
Gender	Man	0.0037
	Woman	-0.0054
Race	African American	-0.1022***
	Asian	0.0661***
	Latino	0.0344*
	Multiracial	0.0025
	White	-0.0028

4 Results

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided SDB dimensions can partially explain polarization in the toxicity/racism annotations. The results for the Kumar et al. [10] and Sap et al. [16] datasets can be found in Tables 4 and 5 respectively, while Tables 2 and 3 display the results for the DICES-350 and DICES-990 datasets respectively.

We observe statistically significant polarization effects caused by annotator *Ethnicity* on all datasets In the DICES dataset, non-white annotators seem to contribute to polarization much more than the majority white annotators. This effect is reversed for the Sap et al. [17] dataset.

Annotator *age* does not cause polarization for toxicity/hate detection, but does for safety None of the age groups in the Sap et al. [17] and Kumar et al. [10] datasets produce statistically significant results for polarization attribution, suggesting that it is not an important factor. However, both DICES datasets produced statistically and quantita-

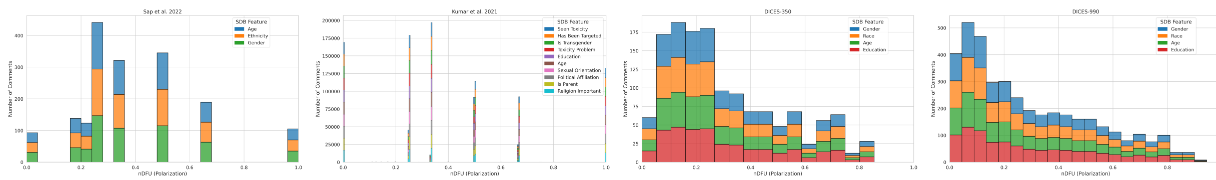

Figure 5: Histogram of dataset polarization for the human datasets considered in this paper per SDB feature.

Table 3: Aposteriori unimodality results for the dices-990 dataset.

SDB Feature		apunim
Age	1) Gen. X+	-0.0639***
	2) Millenial	-0.0133
	3) Gen. Z	0.1076***
Education	College degree or higher	0.0144
	High school or below	-0.0843***
Gender	Man	0.0222*
	Nonbinary	—
	Other	—
	Woman	-0.0157
Race	African American	0.1093***
	Asian	0.0555***
	Latino	-0.1962***
	Other	0.0673***
	White	-0.1480***

Table 4: Aposteriori unimodality results for the sap dataset.

SDB Feature		apunim
Age	0-20	—
	20-40	0.0017
	40-60	0.0115
	60-80	0.0049
Ethnicity	black	-0.2967***
	hisp	—
	middleEastern	—
	other	—
	white	0.1400***
Gender	man	0.0942***
	nonBinary	—
	woman	-0.1810***

tively significant results. This may be caused by the large number of annotators in these datasets, which enable our method to better distinguish this effect from noise in the annotations. This is unlikely however, given our next finding.

Polarization attribution follows ordinal ranking While most SDB attributes are categorical, some are ordinal (e.g., Age, Education, LIKERT-scale opinions). There is a persistent pattern among (statistically significant ordinal attributions), where polarization changes monotonically with their ranking. Examples include:

- Polarization increasing/decreasing with annotator age in both DICES datasets (but not in the Sap et al. [17] and Kumar et al. [10] datasets, where there is no statistical significance).
- Thoughts on religion and toxicity in the Kumar et al. [10] dataset (although much more noisily, since some attribute factors do not display statistical significance).

A notable exception is *Education* in the Kumar et al. [10] dataset. This could be attributed to the curse of dimensionality, since we investigate a great number of groups, but there are few annotators for each individual group.

The fact that this behavior is not replicated when investigating annotator age in the Sap et al. [17] and Kumar et al. [10] datasets lends further support to the argument that age does indeed not play a major role in annotator polarization for these datasets.

5 Conclusion

We introduced the "Aposteriori Unimodality Test", a statistical test that detects whether certain sociodemographic groups partly cause polarization in annotations. Our test is resistant to low samples of annotations, bypasses common statistical issues such as sample dependence, and avoids newly-discovered issues such as the presence of inherent polarization and polarization direction. We apply our test two human-generated and two LLM-generated datasets and find interesting patterns on the former, and verify current literature on LLM SDB prompting on the latter.

6 Limitations

Since there are no other established tests that attribute polarization to sociodemographic characteristics, there is no way to verify that the results presented in §4 are accurate. Similarly, there is no qualitative way of evaluating our test, other than observations on the (non-) efficacy of LLM SDB prompting and our own intuition. Lastly, while our test seems to work sufficiently well with a small number of annotators per data-point, we still encourage researchers using this test to have at least a few annotators of each sociodemographic group that is being tested, in their dataset.

7 Ethical Considerations

While our works aims to help protect marginalized and disadvantaged groups by attributing polarization to certain subgroups, it can be taken advantage of by malicious actors. Given a dataset with fine-grained SDB information, these actors can use our test to target specific vulnerable groups. We thus urge researchers to keep datasets including such information protected, and provided to others only under explicit permission.

Table 5: Aposteriori unimodality results for the kumar dataset.

SDB Feature		apunim
Age	18 - 24	-0.0081
	25 - 34	0.0220
	35 - 44	-0.0176
	45 - 54	0.0079
	55 - 64	-0.0077
	65 or older	-0.0022
	Prefer not to say	—
	Under 18	—
Education	1) Doctoral degree	0.0047
	2) Master's degree	0.0196
	3) Professional degree	-0.0001
	4) Bachelor's degree	0.0365^{***}
	5) Associate degree	-0.0066
	6) College, no degree	-0.0299^{**}
	7) High School graduate	-0.0146
	8) No high school	-0.0059
	Other	—
	Prefer not to say	—
	NaN	—
Has Been Targeted	False	-0.0521^{***}
	True	0.0222^{*}
Is Parent	No	-0.0677^{***}
	Prefer not to say	0.0051
	Yes	0.0513^{***}
Is Transgender	No	-0.0703^{***}
	Prefer not to say	0.0036
	Yes	-0.0027
Political Affiliation	Conservative	0.0287^{**}
	Independent	-0.0159
	Liberal	-0.0258^{*}
	Other	-0.0085
	Prefer not to say	0.0040
Religion Important	1) No	-0.0615^{***}
	2) Not very	-0.0126
	3) Somewhat	0.0095
	4) Very	0.0595^{***}
	Prefer not to say	0.0005
Seen Toxicity	False	0.0386^{***}
	True	-0.0751^{***}
Sexual Orientation	Bisexual	0.0101
	Heterosexual	-0.0513^{***}
	Homosexual	-0.0048
	Other	0.0006
	Prefer not to say	-0.0053
	NaN	0.0093
Toxicity Problem	1) Never	0.0029
	2) Rarely	0.0142
	3) Occasionally	-0.0033
	4) Frequently	-0.0300^{**}
	5) Very Frequently	-0.0204^{*}

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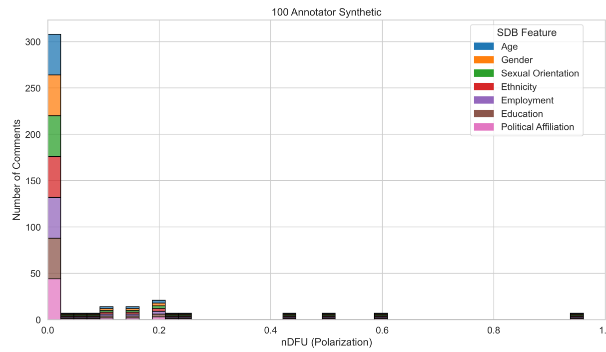

Figure 6: Histogram of dataset polarization for the synthetic datasets considered in this paper per SDB feature.

A Acronyms

LLM Large Language Model

SDB SocioDemographic Background

nDFU normalized Distance From Unimodality

FWER Family-Wise Error Rate

B LLM Annotation

An issue with synthetic datasets is the lack of overall polarization (see Figure 6). This can be offset by the ability of synthetic datasets to be scaled to a much greater extent than human datasets. This is because LLMs do not have cost and availability constraints of humans.

The results for the 100-annotator dataset can be found in Table **??**. The lack of overall polarization, as well as the quantitatively smaller apunim values, suggest that SDB prompting can not be used to simulate annotations for human groups. This is consistent with relevant literature on the topic [1, 6, 15, 8, 3, 11].

C Effect of annotation size

While our test works best with many annotations per datapoint, the cost required for multiple human an-

Table 6: Aposteriori unimodality results for the 100 dataset.

SDB Feature		apunim
Age	0-20	0.0447*
	20-40	-0.0310
	40-60	-0.0829*
	60-80	0.0415
Education	bachelor's degree	-0.0649*
	doctoral degree	-0.0107
	master's degree	-0.0155
	no formal education	-0.0596*
	primary education	0.0275
	secondary education	0.0197
	vocational training	0.0007
Gender	female	0.0095
	male	-0.0144
	non-binary	0.0298
Politics	apolitical	-0.0381
	centrist	-0.0100
	left-wing	-0.0205
	right-wing	0.0044
Sexual Orientation	asexual	0.0031
	bisexual	-0.0338
	heterosexual	-0.0037
	homosexual	0.0319

notators is often prohibitive [15]. Figure 7 demonstrates the effect of the number of annotators on the reliability of the polarization estimation. We can see that the metric becomes consistent with about 15 annotators per comment, after which we start to get diminishing results. This experiment was conducted on a subset of the Sap et al. [16] dataset, with LLM annotators having different SDBs. The full experimental procedure can be found in Appendix B.

D Theoretical limitations

In §2.2 we compared the polarization of the whole set of annotations with the partitions by annotator group. This approach uses the same samples for both subset and superset, violating the fundamental hy-

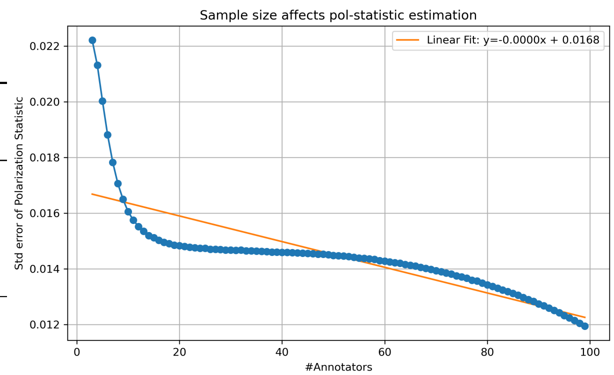

Figure 7: Standard error of the *pol-statistic* when sampled with replacement from the 100 LLM-annotator synthetic dataset for various number of annotators.

pothesis of independent samples. This also means the statistics for both will *always be similar to an extent*. This is especially pronounced in our case, due to the extreme overlap exhibited by tiny annotation sets. Additionally, when calculating p-values, the non-independence of samples induces an intrinsic, non-zero covariance between any statistic s calculated on A and on S.

A standard method of side-stepping this issue is to compare the subset with its complement to the superset. This can not be applied to our case. Consider the example in Figure 2; if we compared $nDFU_{men}$ with $nDFU_{women}$, the result would be very close to zero, indicating no polarization attribution to gender.

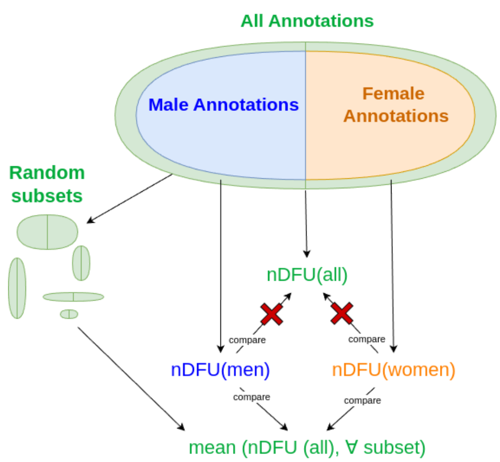

Figure 8: Since we can not compare the statistics of the group-subsets with the annotation superset, we elect to instead compare them with the mean apriori polarization of many sampled subsets.